T3 intermediate-EFT one-loop RGE term comparison: EFT_1 after F

This stage has active field content SM+$S_1 + S_2$. The report combines the renormalisable RGBeta running with the dimension-five $\psi^2\phi^2$ Wilson-coefficient running generated after integrating out F. Throughout, $16\pi^2\,\mu\,dX/d\mu = \beta_X^{(1)}$. For the Wilson sector only the tree-generated stage-1 coefficient $C^{(0)}$ is inserted into the one-loop anomalous dimension; the one-loop threshold piece is retained as a boundary term and is not run again at this order.

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset i g_1 Y B_\mu$
g_2	$D_\mu \supset i g_2 T^A W_\mu^A$
g_3	$D_\mu \supset i g_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \widetilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$L F S_1 + \text{h.c.}$
y_2	$L F S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\mathrm{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\mathrm{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\mathrm{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$

$C_2(d)$	$(d^2-1)/4$
$T(d)$	$d(d^2-1)/12$
$C^{ij;AB}_{LLS_1S_2}$	$L_iL_jS_1^AS_2^B$
C_{ijab}	$L_iL_j\phi_a\phi_b$
C_5^{ij}	$(L_i^TCL_j)HH$
R_W	$16\pi^2\beta_{C_5} \supset R_W\lambda_5 C_{LLS_1S_2}$

Renormalisable EFT1 couplings

$\beta^{(1)}_{g_1}$

Quantity: g_1

$$16\pi^2\,\mu\frac{dg_1}{d\mu} = \beta^{(1)}_{g_1}$$

Column	Representative term structure
T1	$\frac{43{g_1}^3}{6}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{43{g_1}^3}{6}$

$\beta_{g_2}^{(1)}$

Quantity: g_2

$$16\pi^2\,\mu\frac{dg_2}{d\mu} = \beta_{g_2}^{(1)}$$

Column	Representative term structure
T1	$-\frac{17{g_2}^3}{6}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{17{g_2}^3}{6}$

$\beta_{g_3}^{(1)}$

Quantity: g_3

$$16\pi^2\,\mu\frac{dg_3}{d\mu} = \beta_{g_3}^{(1)}$$

Column	Representative term structure
T1	$-7{g_3}^3$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-7{g_3}^3$

$\beta^{(1)}_{Y_u}$

Quantity: Y_u

$$16\pi^2\,\mu\frac{dY_u}{d\mu} = \beta^{(1)}_{Y_u}$$

Terms 1 of 3

Column	Representative term structure
T1	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$
T2	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$
T3	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$

Quantity: Y_u

Terms 2 of 3

Column	Representative term structure
T1	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$
T2	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$
T3	$-8{g_3}^2\ (Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$	$-8{g_3}^2\ (Y_u)_{ij}$

Quantity: Y_u

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$

$\beta^{(1)}_{Y_d}$

Quantity: Y_d

$$16\pi^2\,\mu\frac{dY_d}{d\mu} = \beta^{(1)}_{Y_d}$$

Terms 1 of 3

Column	Representative term structure
T1	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$
T2	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$
T3	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$	$+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$

Quantity: Y_d

Terms 2 of 3

Column	Representative term structure
T1	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$
T2	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$
T3	$-8{g_3}^2\ (Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$	$-8{g_3}^2\ (Y_d)_{ij}$

Quantity: Y_d

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{5}{12}{g_1}^2 (Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{5}{12}{g_1}^2 (Y_d)_{ij}$

$\beta_{Y_e}^{(1)}$

Quantity: Y_e

$$16\pi^2\,\mu\frac{dY_e}{d\mu} = \beta_{Y_e}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$
T2	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$
T3	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$	$+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$	$+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$

Quantity: Y_e

Terms 2 of 2

Column	Representative term structure
T1	$+\frac{3}{2}\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij}$
T2	$-\frac{9}{4}{g_2}^2\ (Y_e)_{ij}$
T3	$-\frac{15}{4}{g_1}^2\ (Y_e)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{3}{2}\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij}$	$-\frac{9}{4}{g_2}^2\ (Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\ (Y_e)_{ij}$

$\beta^{(1)}_{m_1^2}$

Quantity: m_1^2

$$16\pi^2\,\mu\frac{dm_1^2}{d\mu} = \beta^{(1)}_{m_1^2}$$

Terms 1 of 2

Column	Representative term structure
T1	$-\frac{9{g_2}^2 m_1^2}{2}$
T2	$-\frac{3{g_1}^2 m_1^2}{2}$
T3	$+4\lambda_{12}^{(1)} m_2^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{9{g_2}^2 m_1^2}{2}$	$-\frac{3{g_1}^2 m_1^2}{2}$	$+4\lambda_{12}^{(1)} m_2^2$

Quantity: m_1^2

Terms 2 of 2

Column	Representative term structure
T1	$+6\lambda_{S_1}^{(1)} m_1^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_{S_1}^{(1)} m_1^2$

$\beta^{(1)}_{m_2^2}$

Quantity: m_2^2

$$16\pi^2\,\mu\frac{dm_2^2}{d\mu} = \beta^{(1)}_{m_2^2}$$

Terms 1 of 2

Column	Representative term structure
T1	$-\frac{9{g_2}^2 m_2^2}{2}$
T2	$-\frac{3{g_1}^2 m_2^2}{2}$
T3	$+4\lambda_{12}^{(1)} m_1^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{9{g_2}^2 m_2^2}{2}$	$-\frac{3{g_1}^2 m_2^2}{2}$	$+4\lambda_{12}^{(1)} m_1^2$

Quantity: m_2^2

Terms 2 of 2

Column	Representative term structure
T1	$+6\lambda_{S_2}^{(1)} m_2^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_{S_2}^{(1)} m_2^2$

$\beta^{(1)}_{\lambda_1}$

Quantity: λ_1

$$16\pi^2\,\mu\frac{d\lambda_1}{d\mu} = \beta^{(1)}_{\lambda_1}$$

Terms 1 of 6

Column	Representative term structure
T1	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$
T3	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$	$-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$

Quantity: λ_1

Terms 2 of 6

Column	Representative term structure
T1	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$
T2	$+8\lambda_7\ (\lambda_7)^*$
T3	$+4\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$	$+8\lambda_7\ (\lambda_7)^*$	$+4\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$

Quantity: λ_1

Terms 3 of 6

Column	Representative term structure
T1	$+8\lambda_8 \left(\lambda_8\right)^*$
T2	$+4\lambda_5 \left(\lambda_5\right)^*$
T3	$+\frac{9{g_2}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_8 \left(\lambda_8\right)^*$	$+4\lambda_5 \left(\lambda_5\right)^*$	$+\frac{9{g_2}^4}{4}$

Quantity: λ_1

Terms 4 of 6

Column	Representative term structure
T1	$+{g_1}^2\left(\frac{{3g_2}^2}{2}-3\lambda_1\right)$
T2	$-9{g_2}^2\lambda_1$
T3	$+\frac{3{g_1}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+{g_1}^2\left(\frac{{3g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$	$+\frac{3{g_1}^4}{4}$

Quantity: λ_1

Terms 5 of 6

Column	Representative term structure
T1	$+12{\lambda_1}^2$
T2	$+4{\lambda_{HS_1}^{(1)}}^2$
T3	$+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+12{\lambda_1}^2$	$+4{\lambda_{HS_1}^{(1)}}^2$	$+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$

Quantity: λ_1

Terms 6 of 6

Column	Representative term structure
T1	$+4\lambda_{HS_2}^{(1)\ 2}$
T2	$+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{HS_2}^{(1)\ 2}$	$+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$

$\beta^{(1)}_{\lambda_{S_1}^{(1)}}$

Quantity: $\lambda_{S_1}^{(1)}$

$$16\pi^2\,\mu\frac{d\lambda_{S_1}^{(1)}}{d\mu} = \beta^{(1)}_{\lambda_{S_1}^{(1)}}$$

Terms 1 of 4

Column	Representative term structure
T1	$8\lambda_7\,(\lambda_7)^*$
T2	$+8\lambda_9\,(\lambda_9)^*$
T3	$+24\lambda_{11}\,(\lambda_{11})^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$8\lambda_7\,(\lambda_7)^*$	$+8\lambda_9\,(\lambda_9)^*$	$+24\lambda_{11}\,(\lambda_{11})^*$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 2 of 4

Column	Representative term structure
T1	$+\frac{9{g_2}^4}{4}$
T2	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_1}^{(1)}\right)$
T3	$-9{g_2}^2\lambda_{S_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_1}^{(1)}\right)$	$-9{g_2}^2\lambda_{S_1}^{(1)}$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 3 of 4

Column	Representative term structure
T1	$+\frac{3{g_1}^4}{4}$
T2	$+4{\lambda_{12}^{(1)}}^2$
T3	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{3{g_1}^4}{4}$	$+4{\lambda_{12}^{(1)}}^2$	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 4 of 4

Column	Representative term structure
T1	$+4{\lambda_{HS_1}^{(1)}}^2$
T2	$+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$
T3	$+12{\lambda_{S_1}^{(1)}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4{\lambda_{HS_1}^{(1)}}^2$	$+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$	$+12{\lambda_{S_1}^{(1)}}^2$

$\beta^{(1)}_{\lambda^{(1)}_{S_2}}$

Quantity: $\lambda^{(1)}_{S_2}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{S_2}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{S_2}}$$

Terms 1 of 4

Column	Representative term structure
T1	$8\lambda_8\,(\lambda_8)^*$
T2	$+8\lambda_9\,(\lambda_9)^*$
T3	$+24\lambda_{10}\,(\lambda_{10})^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$8\lambda_8\,(\lambda_8)^*$	$+8\lambda_9\,(\lambda_9)^*$	$+24\lambda_{10}\,(\lambda_{10})^*$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 2 of 4

Column	Representative term structure
T1	$+\frac{9{g_2}^4}{4}$
T2	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_2}^{(1)}\right)$
T3	$-9{g_2}^2\lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{9{g_2}^4}{4}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_2}^{(1)}\right)$	$-9{g_2}^2\lambda_{S_2}^{(1)}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 3 of 4

Column	Representative term structure
T1	$+\frac{3{g_1}^4}{4}$
T2	$+4{\lambda_{12}^{(1)}}^2$
T3	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{3{g_1}^4}{4}$	$+4{\lambda_{12}^{(1)}}^2$	$+\frac{{\lambda_{12}^{(A)}}^2}{4}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 4 of 4

Column	Representative term structure
T1	$+4{\lambda_{HS_2}^{(1)}}^2$
T2	$+\frac{{\lambda_{HS_2}^{(A)}}^2}{4}$
T3	$+12{\lambda_{S_2}^{(1)}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4{\lambda_{HS_2}^{(1)}}^2$	$+\frac{{\lambda_{HS_2}^{(A)}}^2}{4}$	$+12{\lambda_{S_2}^{(1)}}^2$

$\beta^{(1)}_{\lambda^{(1)}_{HS_1}}$

Quantity: $\lambda^{(1)}_{HS_1}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{HS_1}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{HS_1}}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda^{(1)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$+24\lambda_7\,(\lambda_7)^*$
T3	$+3\lambda_{11}\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda^{(1)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+24\lambda_7\,(\lambda_7)^*$	$+3\lambda_{11}\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$

Quantity: $\lambda^{(1)}_{HS_1}$

Terms 2 of 5

Column	Representative term structure
T1	$+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + 3\left(\lambda_{11}\right)^*)$
T2	$+6\lambda_5\left(\lambda_5\right)^*$
T3	$+\frac{{9g_2}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + 3\left(\lambda_{11}\right)^*)$	$+6\lambda_5\left(\lambda_5\right)^*$	$+\frac{{9g_2}^4}{4}$

Quantity: $\lambda^{(1)}_{HS_1}$

Terms 3 of 5

Column	Representative term structure
T1	$-9{g_2}^2\lambda^{(1)}_{HS_1}$
T2	$+\frac{3{g_1}^4}{4}$
T3	$-3{g_1}^2\lambda^{(1)}_{HS_1}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9{g_2}^2\lambda^{(1)}_{HS_1}$	$+\frac{3{g_1}^4}{4}$	$-3{g_1}^2\lambda^{(1)}_{HS_1}$

Quantity: $\lambda^{(1)}_{HS_1}$

Terms 4 of 5

Column	Representative term structure
T1	$+4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$
T2	$+6\lambda_1\lambda^{(1)}_{HS_1}$
T3	$+4{\lambda^{(1)}_{HS_1}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$	$+6\lambda_1\lambda^{(1)}_{HS_1}$	$+4{\lambda^{(1)}_{HS_1}}^2$

Quantity: $\lambda^{(1)}_{HS_1}$

Terms 5 of 5

Column	Representative term structure
T1	$+6\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$
T2	$+\frac{3\lambda^{(A)\ 2}_{HS_1}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	$+\frac{3\lambda^{(A)\ 2}_{HS_1}}{4}$

$\beta^{(1)}_{\lambda^{(1)}_{HS_2}}$

Quantity: $\lambda^{(1)}_{HS_2}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{HS_2}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{HS_2}}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$+3\lambda_{10}\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$
T3	$+\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + 3\,(\lambda_{10})^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+3\lambda_{10}\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$	$+\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + 3\,(\lambda_{10})^*)$

Quantity: $\lambda^{(1)}_{HS_2}$

Terms 2 of 5

Column	Representative term structure
T1	$+24\lambda_8 \left(\lambda_8\right)^*$
T2	$+6\lambda_5 \left(\lambda_5\right)^*$
T3	$+\frac{9{g_2}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+24\lambda_8 \left(\lambda_8\right)^*$	$+6\lambda_5 \left(\lambda_5\right)^*$	$+\frac{9{g_2}^4}{4}$

Quantity: $\lambda^{(1)}_{HS_2}$

Terms 3 of 5

Column	Representative term structure
T1	$-9{g_2}^2\lambda^{(1)}_{HS_2}$
T2	$+\frac{3{g_1}^4}{4}$
T3	$-3{g_1}^2\lambda^{(1)}_{HS_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9{g_2}^2\lambda^{(1)}_{HS_2}$	$+\frac{3{g_1}^4}{4}$	$-3{g_1}^2\lambda^{(1)}_{HS_2}$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 4 of 5

Column	Representative term structure
T1	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$
T2	$+6\lambda_1\lambda_{HS_2}^{(1)}$
T3	$+4{\lambda_{HS_2}^{(1)}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4{\lambda_{HS_2}^{(1)}}^2$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 5 of 5

Column	Representative term structure
T1	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$
T2	$+\frac{3\lambda_{HS_2}^{(A)\ 2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+\frac{3\lambda_{HS_2}^{(A)\ 2}}{4}$

$\beta^{(1)}_{\lambda^{(1)}_{12}}$

Quantity: $\lambda^{(1)}_{12}$

$$16\pi^2 \, \mu \frac{d\lambda^{(1)}_{12}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{12}}$$

Terms 1 of 5

Column	Representative term structure
T1	$2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} \left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$
T2	$+24\lambda_9 \, (\lambda_9)^*$
T3	$+9\lambda_{10}((\lambda_{10})^* + (\lambda_{11})^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} \left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$	$+24\lambda_9 \, (\lambda_9)^*$	$+9\lambda_{10}((\lambda_{10})^* + (\lambda_{11})^*)$ $+9\lambda_{11}((\lambda_{10})^* + (\lambda_{11})^*)$

Quantity: $\lambda_{12}^{(1)}$

Terms 2 of 5

Column	Representative term structure
T1	$+3\lambda_5 \left(\lambda_5\right)^*$
T2	$+\frac{9{g_2}^4}{4}$
T3	$-9{g_2}^2\lambda_{12}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+3\lambda_5 \left(\lambda_5\right)^*$	$+\frac{9{g_2}^4}{4}$	$-9{g_2}^2\lambda_{12}^{(1)}$

Quantity: $\lambda_{12}^{(1)}$

Terms 3 of 5

Column	Representative term structure
T1	$+\frac{3{g_1}^4}{4}$
T2	$-3{g_1}^2\lambda_{12}^{(1)}$
T3	$+4{\lambda_{12}^{(1)}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{3{g_1}^4}{4}$	$-3{g_1}^2\lambda_{12}^{(1)}$	$+4{\lambda_{12}^{(1)}}^2$

Quantity: $\lambda_{12}^{(1)}$

Terms 4 of 5

Column	Representative term structure
T1	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
T2	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
T3	$+\frac{3{\lambda_{12}^{(A)}}^2}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+\frac{3{\lambda_{12}^{(A)}}^2}{4}$

Quantity: $\lambda_{12}^{(1)}$

Terms 5 of 5

Column	Representative term structure
T1	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$

$\beta^{(1)}_{\lambda_5}$

Quantity: λ_5

$$16\pi^2\,\mu\frac{d\lambda_5}{d\mu} = \beta^{(1)}_{\lambda_5}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_5\left(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + \frac{\lambda^{(A)}_{12}}{6} + \frac{2\lambda^{(A)}_{HS_1}}{3} + 2\lambda^{(A)}_{HS_2}\right)$
T2	$-4\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\,(\lambda_7)^*$
T3	$+4\lambda_{11}\,(\lambda_7)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_5\left(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + \frac{\lambda^{(A)}_{12}}{6} + \frac{2\lambda^{(A)}_{HS_1}}{3} + 2\lambda^{(A)}_{HS_2}\right)$	$-4\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\,(\lambda_7)^*$	$+4\lambda_{11}\,(\lambda_7)^*$

Quantity: λ_5

Terms 2 of 4

Column	Representative term structure
T1	$+4\,(\lambda_8)^*\,((\lambda_{10})^* - \left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*)$
T2	$-9{g_2}^2\lambda_5$
T3	$-3{g_1}^2\lambda_5$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\,(\lambda_8)^*\,((\lambda_{10})^* - \left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*)$	$-9{g_2}^2\lambda_5$	$-3{g_1}^2\lambda_5$

Quantity: λ_5

Terms 3 of 4

Column	Representative term structure
T1	$+2\lambda_{12}^{(1)}\lambda_5$
T2	$+2\lambda_1\lambda_5$
T3	$+4\lambda_{HS_1}^{(1)}\lambda_5$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$

Quantity: λ_5

Terms 4 of 4

Column	Representative term structure
T1	$+4\lambda_{HS_2}^{(1)}\lambda_5$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{HS_2}^{(1)}\lambda_5$

$\beta^{(1)}_{\lambda_{12}^{(A)}}$

Quantity: $\lambda_{12}^{(A)}$

$$16\pi^2\,\mu\frac{d\lambda_{12}^{(A)}}{d\mu} = \beta^{(1)}_{\lambda_{12}^{(A)}}$$

Terms 1 of 4

Column	Representative term structure
T1	$\frac{64}{3}\lambda_9\,(\lambda_9)^*$
T2	$+\frac{4}{3}\lambda_{10}(5\,(\lambda_{10})^* + (\lambda_{11})^*)$
T3	$+\frac{4}{3}\lambda_{11}((\lambda_{10})^* + 5\,(\lambda_{11})^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{64}{3}\lambda_9\,(\lambda_9)^*$	$+\frac{4}{3}\lambda_{10}(5\,(\lambda_{10})^* + (\lambda_{11})^*)$	$+\frac{4}{3}\lambda_{11}((\lambda_{10})^* + 5\,(\lambda_{11})^*)$

Quantity: $\lambda_{12}^{(A)}$

Terms 2 of 4

Column	Representative term structure
T1	$+\frac{4}{3}\lambda_5 \left(\lambda_5\right)^*$
T2	$+{g_1}^2 \left(-6{g_2}^2 - 3\lambda_{12}^{(A)}\right)$
T3	$-9{g_2}^2\lambda_{12}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{4}{3}\lambda_5 \left(\lambda_5\right)^*$	$+{g_1}^2 \left(-6{g_2}^2 - 3\lambda_{12}^{(A)}\right)$	$-9{g_2}^2\lambda_{12}^{(A)}$

Quantity: $\lambda_{12}^{(A)}$

Terms 3 of 4

Column	Representative term structure
T1	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
T2	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$
T3	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$

Quantity: $\lambda_{12}^{(A)}$

Terms 4 of 4

Column	Representative term structure
T1	$+\lambda_{HS_1}^{(A)}\,\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{HS_1}^{(A)}\,\lambda_{HS_2}^{(A)}$

$\beta^{(1)}_{\lambda^{(A)}_{HS_1}}$

Quantity: $\lambda^{(A)}_{HS_1}$

$$16\pi^2\,\mu\frac{d\lambda^{(A)}_{HS_1}}{d\mu} = \beta^{(1)}_{\lambda^{(A)}_{HS_1}}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda^{(A)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$+\frac{64}{3}\lambda_7\,(\lambda_7)^*$
T3	$-\frac{4}{3}\lambda_{11}\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda^{(A)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+\frac{64}{3}\lambda_7\,(\lambda_7)^*$	$-\frac{4}{3}\lambda_{11}\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$

Quantity: $\lambda^{(A)}_{HS_1}$

Terms 2 of 4

Column	Representative term structure
T1	$-\frac{4}{3}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + (\lambda_{11})^*)$
T2	$+\frac{16}{3}\lambda_5\,(\lambda_5)^*$
T3	$+{g_1}^2\left(-6{g_2}^2 - 3\lambda^{(A)}_{HS_1}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{4}{3}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + (\lambda_{11})^*)$	$+\frac{16}{3}\lambda_5\,(\lambda_5)^*$	$+{g_1}^2\left(-6{g_2}^2 - 3\lambda^{(A)}_{HS_1}\right)$

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 3 of 4

Column	Representative term structure
T1	$-9{g_2}^2\lambda_{HS_1}^{(A)}$
T2	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
T3	$+2\lambda_1\lambda_{HS_1}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9{g_2}^2\lambda_{HS_1}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$	$+2\lambda_1\lambda_{HS_1}^{(A)}$

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 4 of 4

Column	Representative term structure
T1	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$
T2	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$

$\beta^{(1)}_{\lambda^{(A)}_{HS_2}}$

Quantity: $\lambda^{(A)}_{HS_2}$

$$16\pi^2\,\mu\frac{d\lambda^{(A)}_{HS_2}}{d\mu} = \beta^{(1)}_{\lambda^{(A)}_{HS_2}}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda^{(A)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$-4\lambda_{10}\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$
T3	$-4\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + (\lambda_{10})^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda^{(A)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-4\lambda_{10}\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$	$-4\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + (\lambda_{10})^*)$

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 2 of 4

Column	Representative term structure
T1	$+64\lambda_8 \left(\lambda_8\right)^*$
T2	$+16\lambda_5 \left(\lambda_5\right)^*$
T3	$+{g_1}^2 \left(6{g_2}^2 - 3\lambda_{HS_2}^{(A)}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+64\lambda_8 \left(\lambda_8\right)^*$	$+16\lambda_5 \left(\lambda_5\right)^*$	$+{g_1}^2 \left(6{g_2}^2 - 3\lambda_{HS_2}^{(A)}\right)$

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 3 of 4

Column	Representative term structure
T1	$-9{g_2}^2\lambda_{HS_2}^{(A)}$
T2	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
T3	$+2\lambda_1\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9{g_2}^2\lambda_{HS_2}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$

Quantity: $\lambda^{(A)}_{HS_2}$

Terms 4 of 4

Column	Representative term structure
T1	$+8\lambda^{(1)}_{HS_2}\lambda^{(A)}_{HS_2}$
T2	$+2\lambda^{(A)}_{HS_2}\lambda^{(1)}_{S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda^{(1)}_{HS_2}\lambda^{(A)}_{HS_2}$	$+2\lambda^{(A)}_{HS_2}\lambda^{(1)}_{S_2}$

$\beta^{(1)}_{\lambda_7}$

Quantity: λ_7

$$16\pi^2\,\mu\frac{d\lambda_7}{d\mu} = \beta^{(1)}_{\lambda_7}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_7(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$-2\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\,(\lambda_5)^*$
T3	$+2\lambda_{11}\,(\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_7(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-2\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\,(\lambda_5)^*$	$+2\lambda_{11}\,(\lambda_5)^*$

Quantity: λ_7

Terms 2 of 4

Column	Representative term structure
T1	$-9{g_2}^2\lambda_7$
T2	$-3{g_1}^2\lambda_7$
T3	$+2\lambda_1\lambda_7$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9{g_2}^2\lambda_7$	$-3{g_1}^2\lambda_7$	$+2\lambda_1\lambda_7$

Quantity: λ_7

Terms 3 of 4

Column	Representative term structure
T1	$+8\lambda_{HS_1}^{(1)}\lambda_7$
T2	$+\frac{4\lambda_{HS_1}^{(A)}\lambda_7}{3}$
T3	$+2\lambda_7\lambda_{S_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_{HS_1}^{(1)}\lambda_7$	$+\frac{4\lambda_{HS_1}^{(A)}\lambda_7}{3}$	$+2\lambda_7\lambda_{S_1}^{(1)}$

Quantity: λ_7

Terms 4 of 4

Column	Representative term structure
T1	$+4\lambda_8\lambda_9$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_8\lambda_9$

$\beta^{(1)}_{\lambda_{12}}$

Quantity: λ_{12}

$$16\pi^2\,\mu\frac{d\lambda_{12}}{d\mu} = \beta^{(1)}_{\lambda_{12}}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$+6\lambda_9\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$
T3	$-6\,(\lambda_8)^*\,(\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+6\lambda_9\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$	$-6\,(\lambda_8)^*\,(\lambda_5)^*$

Quantity: λ_{12}

Terms 2 of 4

Column	Representative term structure
T1	$-9{g_2}^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$
T2	$-3{g_1}^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$
T3	$+2\lambda_{12}^{(1)}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9{g_2}^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	$-3{g_1}^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	$+2\lambda_{12}^{(1)}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$

Quantity: λ_{12}

Terms 3 of 4

Column	Representative term structure
T1	$+3\lambda_1\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$
T2	$+\lambda^{(1)}_{HS_1}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+3\lambda_{11})$
T3	$+\lambda^{(A)}_{HS_1}\left(-\frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}-\frac{\lambda_{11}}{4}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+3\lambda_1\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	$+\lambda^{(1)}_{HS_1}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+3\lambda_{11})$	$+\lambda^{(A)}_{HS_1}\left(-\frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}-\frac{\lambda_{11}}{4}\right)$

Quantity: λ_{12}

Terms 4 of 4

Column	Representative term structure
T1	$+\lambda^{(1)}_{HS_2}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+3\lambda_{10})$
T2	$+\lambda^{(A)}_{HS_2}\left(-\frac{3\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}-\frac{3\lambda_{10}}{4}\right)$
T3	$-6\lambda_7\lambda_5$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda^{(1)}_{HS_2}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+3\lambda_{10})$	$+\lambda^{(A)}_{HS_2}\left(-\frac{3\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}-\frac{3\lambda_{10}}{4}\right)$	$-6\lambda_7\lambda_5$

$\beta^{(1)}_{\lambda_8}$

Quantity: λ_8

$$16\pi^2\,\mu\frac{d\lambda_8}{d\mu} = \beta^{(1)}_{\lambda_8}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda_8(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T2	$+4\lambda_7\,(\lambda_9)^*$
T3	$+2\,(\lambda_5)^*\,((\lambda_{10})^* - \left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_8(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$+4\lambda_7\,(\lambda_9)^*$	$+2\,(\lambda_5)^*\,((\lambda_{10})^* - \left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*)$

Quantity: λ_8

Terms 2 of 3

Column	Representative term structure
T1	$-9{g_2}^2\lambda_8$
T2	$-3{g_1}^2\lambda_8$
T3	$+2\lambda_1\lambda_8$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9{g_2}^2\lambda_8$	$-3{g_1}^2\lambda_8$	$+2\lambda_1\lambda_8$

Quantity: λ_8

Terms 3 of 3

Column	Representative term structure
T1	$+8\lambda^{(1)}_{HS_2}\lambda_8$
T2	$+4\lambda^{(A)}_{HS_2}\lambda_8$
T3	$+2\lambda_8\lambda^{(1)}_{S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda^{(1)}_{HS_2}\lambda_8$	$+4\lambda^{(A)}_{HS_2}\lambda_8$	$+2\lambda_8\lambda^{(1)}_{S_2}$

$\beta^{(1)}_{\lambda_{11}}$

Quantity: λ_{11}

$$16\pi^2\,\mu\frac{d\lambda_{11}}{d\mu} = \beta^{(1)}_{\lambda_{11}}$$

Terms 1 of 3

Column	Representative term structure
T1	$4\lambda_9((\lambda_{10})^* + 5\,(\lambda_{11})^*)$
T2	$-9{g_2}^2\lambda_{11}$
T3	$-3{g_1}^2\lambda_{11}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$4\lambda_9((\lambda_{10})^* + 5\,(\lambda_{11})^*)$	$-9{g_2}^2\lambda_{11}$	$-3{g_1}^2\lambda_{11}$

Quantity: λ_{11}

Terms 2 of 3

Column	Representative term structure
T1	$+\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$
T2	$+\lambda_{12}^{(A)}\left(\frac{\lambda_{10}}{6}+\frac{5\lambda_{11}}{6}\right)$
T3	$+2\lambda_{HS_1}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$	$+\lambda_{12}^{(A)}\left(\frac{\lambda_{10}}{6}+\frac{5\lambda_{11}}{6}\right)$	$+2\lambda_{HS_1}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$

Quantity: λ_{11}

Terms 3 of 3

Column	Representative term structure
T1	$-\dfrac{\lambda_{HS_1}^{(A)}\,\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}}{6}$
T2	$+4\lambda_7\lambda_5$
T3	$+12\lambda_{S_1}^{(1)}\lambda_{11}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\dfrac{\lambda_{HS_1}^{(A)}\,\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}}{6}$	$+4\lambda_7\lambda_5$	$+12\lambda_{S_1}^{(1)}\lambda_{11}$

$\beta^{(1)}_{\lambda_9}$

Quantity: λ_9

$$16\pi^2\,\mu\frac{d\lambda_9}{d\mu} = \beta^{(1)}_{\lambda_9}$$

Terms 1 of 4

Column	Representative term structure
T1	$4\lambda_7\,(\lambda_8)^*$
T2	$-9{g_2}^2\lambda_9$
T3	$-3{g_1}^2\lambda_9$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$4\lambda_7\,(\lambda_8)^*$	$-9{g_2}^2\lambda_9$	$-3{g_1}^2\lambda_9$

Quantity: λ_9

Terms 2 of 4

Column	Representative term structure
T1	$+8\lambda_{12}^{(1)}\lambda_9$
T2	$+\frac{4\lambda_{12}^{(A)}\lambda_9}{3}$
T3	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}\,2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_{12}^{(1)}\lambda_9$	$+\frac{4\lambda_{12}^{(A)}\lambda_9}{3}$	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}\,2}$

Quantity: λ_9

Terms 3 of 4

Column	Representative term structure
T1	$+2\lambda_{S_1}^{(1)}\lambda_9$
T2	$+2\lambda_9\lambda_{S_2}^{(1)}$
T3	$+5{\lambda_{10}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{S_1}^{(1)}\lambda_9$	$+2\lambda_9\lambda_{S_2}^{(1)}$	$+5{\lambda_{10}}^2$ $+5{\lambda_{11}}^2$

Quantity: λ_9

Terms 4 of 4

Column	Representative term structure
T1	$+2\lambda_{10}\lambda_{11}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{10}\lambda_{11}$

$\beta^{(1)}_{\lambda_{10}}$

Quantity: λ_{10}

$$16\pi^2\,\mu\frac{d\lambda_{10}}{d\mu} = \beta^{(1)}_{\lambda_{10}}$$

Terms 1 of 3

Column	Representative term structure
T1	$4\,(\lambda_8)^*\,(\lambda_5)^*$
T2	$+4\lambda_9(5\,(\lambda_{10})^* + (\lambda_{11})^*)$
T3	$-9{g_2}^2\lambda_{10}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$4\,(\lambda_8)^*\,(\lambda_5)^*$	$+4\lambda_9(5\,(\lambda_{10})^* + (\lambda_{11})^*)$	$-9{g_2}^2\lambda_{10}$

Quantity: λ_{10}

Terms 2 of 3

Column	Representative term structure
T1	$-3{g_1}^2\lambda_{10}$
T2	$+\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$
T3	$+\lambda_{12}^{(A)}\left(\frac{5\lambda_{10}}{6}+\frac{\lambda_{11}}{6}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-3{g_1}^2\lambda_{10}$	$+\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$	$+\lambda_{12}^{(A)}\left(\frac{5\lambda_{10}}{6}+\frac{\lambda_{11}}{6}\right)$

Quantity: λ_{10}

Terms 3 of 3

Column	Representative term structure
T1	$+2\lambda_{HS_2}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T2	$-\frac{\lambda_{HS_2}^{(A)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$
T3	$+12\lambda_{10}\lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{HS_2}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$-\frac{\lambda_{HS_2}^{(A)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$	$+12\lambda_{10}\lambda_{S_2}^{(1)}$

Dimension-five $\psi^2\phi^2$ Wilson coefficients

The Wilson tensor is reported in the real-scalar component basis C_{ijab}. Each component with a nonzero one-loop beta function has its own table. Components absent at the matching boundary but generated by operator mixing are included automatically.

$\beta^{(1)}_{C_{1133}}$

Quantity: 1, 1, 3, 3

$$16\pi^2\,\mu\frac{dC_{1133}}{d\mu} = \beta^{(1)}_{C_{1133}}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}lambdaT_3\overline{y_1}\,\overline{y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{2\sqrt{3}lambdaT_3\overline{y_1}\,\overline{y_2}}{3MF}$

$\beta^{(1)}_{C_{1134}}$

Quantity: 1, 1, 3, 4

$$16\pi^2\,\mu\frac{dC_{1134}}{d\mu} = \beta^{(1)}_{C_{1134}}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}ilambdaT_3\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{2\sqrt{3}ilambdaT_3\overline{y_1 y_2}}{3MF}$

$\beta^{(1)}_{C_{1144}}$

Quantity: 1, 1, 4, 4

$$16\pi^2\,\mu\,\frac{dC_{1144}}{d\mu} = \beta^{(1)}_{C_{1144}}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}lambdaT_3\overline{y_1}\,\overline{y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1}\,\overline{y_2}}{3MF}$

$\beta^{(1)}_{C_{1155}}$

Quantity: 1, 1, 5, 5

$$16\pi^2\,\mu\frac{dC_{1155}}{d\mu} = \beta^{(1)}_{C_{1155}}$$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1}^2}{MF}$
T2	$\frac{lambdaS1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{ye\overline{y_1}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_2^2\overline{y_1}^2}{4MF}$	$-\frac{lambdaS1Inv1\overline{y_1}^2}{2MF}$	$\frac{ye\overline{y_1}^2\overline{ye}}{4MF}$

$\beta^{(1)}_{C_{1156}}$

Quantity: 1, 1, 5, 6

$$16\pi^2\,\mu\frac{dC_{1156}}{d\mu} = \beta^{(1)}_{C_{1156}}$$

Column	Representative term structure
T1	$\frac{ilambdaS1Inv1\overline{y_1}^2}{MF}$
T2	$\frac{ig_2^2\overline{y_1}^2}{MF}$
T3	$\frac{iye\overline{y_1}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{ilambdaS1Inv1\overline{y_1}^2}{2MF}$	$\frac{3ig_2^2\overline{y_1}^2}{4MF}$	$-\frac{iye\overline{y_1}^2\overline{ye}}{4MF}$

$\beta^{(1)}_{C_{11511}}$

Quantity: 1, 1, 5, 11

$$16\pi^2\,\mu\,\frac{dC_{11511}}{d\mu} = \beta^{(1)}_{C_{11511}}$$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1 y_2}}{MF}$
T2	$\frac{\sqrt{3}lambda12Inv2\overline{y_1 y_2}}{MF}$
T3	$\frac{ye\overline{y_1 y_2 ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_2^2\overline{y_1 y_2}}{4MF}$	$-\frac{\sqrt{3}lambda12Inv2\overline{y_1 y_2}}{3MF}$	$\frac{ye\overline{y_1 y_2 ye}}{4MF}$

$\beta^{(1)}_{C_{11512}}$

Quantity: 1, 1, 5, 12

$$16\pi^2\,\mu\,\frac{dC_{11512}}{d\mu} = \beta^{(1)}_{C_{11512}}$$

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1 y_2}}{MF}$
T2	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1 y_2}}{MF}$
T3	$\frac{iye\overline{y_1 y_2 ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3ig_2^2\overline{y_1 y_2}}{4MF}$	$-\frac{\sqrt{3}ilambda12Inv2\overline{y_1 y_2}}{3MF}$	$\frac{iye\overline{y_1 y_2 ye}}{4MF}$

$\beta^{(1)}_{C_{1166}}$

Quantity: 1, 1, 6, 6

$$16\pi^2\,\mu\frac{dC_{1166}}{d\mu} = \beta^{(1)}_{C_{1166}}$$

Column	Representative term structure
T1	$\frac{lambdaS1Inv1\overline{y_1}^2}{MF}$
T2	$\frac{g_2^2\overline{y_1}^2}{MF}$
T3	$\frac{ye\overline{y_1}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{lambdaS1Inv1\overline{y_1}^2}{2MF}$	$\frac{3g_2^2\overline{y_1}^2}{4MF}$	$-\frac{ye\overline{y_1}^2\overline{ye}}{4MF}$

$\beta^{(1)}_{C_{11611}}$

Quantity: 1, 1, 6, 11

$$16\pi^2\,\mu\,\frac{dC_{11611}}{d\mu} = \beta^{(1)}_{C_{11611}}$$

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1 y_2}}{MF}$
T2	$\frac{iye\overline{y_1 y_2 ye}}{MF}$
T3	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{3ig_2^2\overline{y_1 y_2}}{4MF}$	$-\frac{iye\overline{y_1 y_2 ye}}{4MF}$	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1 y_2}}{3MF}$

$\beta^{(1)}_{C_{11612}}$

Quantity: 1, 1, 6, 12

$$16\pi^2\,\mu\,\frac{dC_{11612}}{d\mu} = \beta^{(1)}_{C_{11612}}$$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1 y_2}}{MF}$
T2	$\frac{\sqrt{3}lambda12Inv2\overline{y_1 y_2}}{MF}$
T3	$\frac{ye\overline{y_1 y_2 ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_2^2\overline{y_1 y_2}}{4MF}$	$-\frac{\sqrt{3}lambda12Inv2\overline{y_1 y_2}}{3MF}$	$\frac{ye\overline{y_1 y_2 ye}}{4MF}$

$\beta^{(1)}_{C_{111111}}$

Quantity: 1, 1, 11, 11

$$16\pi^2\,\mu\frac{dC_{111111}}{d\mu} = \beta^{(1)}_{C_{111111}}$$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_2}^2}{MF}$
T2	$\frac{lambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{ye\overline{y_2}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_2^2\overline{y_2}^2}{4MF}$	$-\frac{lambdaS2Inv1\overline{y_2}^2}{2MF}$	$\frac{ye\overline{y_2}^2\overline{ye}}{4MF}$

$\beta^{(1)}_{C_{111112}}$

Quantity: 1, 1, 11, 12

$$16\pi^2\,\mu\frac{dC_{111112}}{d\mu} = \beta^{(1)}_{C_{111112}}$$

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_2}^2}{MF}$
T2	$\frac{ilambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{iye\overline{y_2}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3ig_2^2\overline{y_2}^2}{4MF}$	$-\frac{ilambdaS2Inv1\overline{y_2}^2}{2MF}$	$\frac{iye\overline{y_2}^2\overline{ye}}{4MF}$

$\beta^{(1)}_{C_{111212}}$

Quantity: 1, 1, 12, 12

$$16\pi^2\,\mu\frac{dC_{111212}}{d\mu} = \beta^{(1)}_{C_{111212}}$$

Column	Representative term structure
T1	$\frac{lambdaS2Inv1\overline{y_2}^2}{MF}$
T2	$\frac{g_2^2\overline{y_2}^2}{MF}$
T3	$\frac{ye\overline{y_2}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{lambdaS2Inv1\overline{y_2}^2}{2MF}$	$\frac{3g_2^2\overline{y_2}^2}{4MF}$	$-\frac{ye\overline{y_2}^2\overline{ye}}{4MF}$

$\beta^{(1)}_{C_{1213}}$

Quantity: 1, 2, 1, 3

$$16\pi^2\,\mu\frac{dC_{1213}}{d\mu} = \beta^{(1)}_{C_{1213}}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}lambdaT_3\overline{y_1}\,\overline{y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{\sqrt{3}lambdaT_3\overline{y_1}\,\overline{y_2}}{3MF}$

$\beta^{(1)}_{C_{1214}}$

Quantity: 1, 2, 1, 4

$$16\pi^2\,\mu\frac{dC_{1214}}{d\mu} = \beta^{(1)}_{C_{1214}}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}ilambdaT_3\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1 y_2}}{3MF}$

$\beta^{(1)}_{C_{1223}}$

Quantity: 1, 2, 2, 3

$$16\pi^2\,\mu\frac{dC_{1223}}{d\mu} = \beta^{(1)}_{C_{1223}}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}ilambdaT_3\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1 y_2}}{3MF}$

$\beta^{(1)}_{C_{1224}}$

Quantity: 1, 2, 2, 4

$$16\pi^2\,\mu\frac{dC_{1224}}{d\mu} = \beta^{(1)}_{C_{1224}}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}lambdaT_3\overline{y_1}\,\overline{y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{\sqrt{3}lambdaT_3\overline{y_1}\,\overline{y_2}}{3MF}$

$\beta^{(1)}_{C_{1257}}$

Quantity: 1, 2, 5, 7

$$16\pi^2\,\mu\frac{dC_{1257}}{d\mu} = \beta^{(1)}_{C_{1257}}$$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1}^2}{MF}$
T2	$\frac{lambdaS1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{ye\overline{y_1}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_2^2\overline{y_1}^2}{8MF}$	$-\frac{lambdaS1Inv1\overline{y_1}^2}{4MF}$	$\frac{ye\overline{y_1}^2\overline{ye}}{8MF}$

$\beta^{(1)}_{C_{1258}}$

Quantity: 1, 2, 5, 8

$$16\pi^2\,\mu\frac{dC_{1258}}{d\mu} = \beta^{(1)}_{C_{1258}}$$

Column	Representative term structure
T1	$\frac{ilambdaS1Inv1\overline{y_1}^2}{MF}$
T2	$\frac{ig_2^2\overline{y_1}^2}{MF}$
T3	$\frac{iye\overline{y_1}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{ilambdaS1Inv1\overline{y_1}^2}{4MF}$	$\frac{3ig_2^2\overline{y_1}^2}{8MF}$	$-\frac{iye\overline{y_1}^2\overline{ye}}{8MF}$

$\beta^{(1)}_{C_{1259}}$

Quantity: 1, 2, 5, 9

$$16\pi^2\,\mu\frac{dC_{1259}}{d\mu} = \beta^{(1)}_{C_{1259}}$$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1 y_2}}{MF}$
T2	$\frac{ye\overline{y_1 y_2 ye}}{MF}$
T3	$\frac{\sqrt{3}lambda12Inv2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{3g_2^2\overline{y_1 y_2}}{8MF}$	$-\frac{ye\overline{y_1 y_2 ye}}{8MF}$	$\frac{\sqrt{3}lambda12Inv2\overline{y_1 y_2}}{6MF}$

$\beta^{(1)}_{C_{12510}}$

Quantity: 1, 2, 5, 10

$$16\pi^2\,\mu\,\frac{dC_{12510}}{d\mu} = \beta^{(1)}_{C_{12510}}$$

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1 y_2}}{MF}$
T2	$\frac{iye\overline{y_1 y_2 ye}}{MF}$
T3	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{3ig_2^2\overline{y_1 y_2}}{8MF}$	$-\frac{iye\overline{y_1 y_2 ye}}{8MF}$	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1 y_2}}{6MF}$

$\beta^{(1)}_{C_{1267}}$

Quantity: 1, 2, 6, 7

$$16\pi^2\,\mu\frac{dC_{1267}}{d\mu} = \beta^{(1)}_{C_{1267}}$$

Column	Representative term structure
T1	$\frac{ilambdaS1Inv1\overline{y_1}^2}{MF}$
T2	$\frac{ig_2^2\overline{y_1}^2}{MF}$
T3	$\frac{iye\overline{y_1}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{ilambdaS1Inv1\overline{y_1}^2}{4MF}$	$\frac{3ig_2^2\overline{y_1}^2}{8MF}$	$-\frac{iye\overline{y_1}^2\overline{ye}}{8MF}$

$\beta^{(1)}_{C_{1268}}$

Quantity: 1, 2, 6, 8

$$16\pi^2\,\mu\frac{dC_{1268}}{d\mu} = \beta^{(1)}_{C_{1268}}$$

Column	Representative term structure
T1	$\frac{lambdaS1Inv1\overline{y_1}^2}{MF}$
T2	$\frac{g_2^2\overline{y_1}^2}{MF}$
T3	$\frac{ye\overline{y_1}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{lambdaS1Inv1\overline{y_1}^2}{4MF}$	$\frac{3g_2^2\overline{y_1}^2}{8MF}$	$-\frac{ye\overline{y_1}^2\overline{ye}}{8MF}$

$\beta^{(1)}_{C_{1269}}$

Quantity: 1, 2, 6, 9

$$16\pi^2\,\mu\frac{dC_{1269}}{d\mu} = \beta^{(1)}_{C_{1269}}$$

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1 y_2}}{MF}$
T2	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1 y_2}}{MF}$
T3	$\frac{iye\overline{y_1 y_2 ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3ig_2^2\overline{y_1 y_2}}{8MF}$	$-\frac{\sqrt{3}ilambda12Inv2\overline{y_1 y_2}}{6MF}$	$\frac{iye\overline{y_1 y_2 ye}}{8MF}$

$\beta^{(1)}_{C_{12610}}$

Quantity: 1, 2, 6, 10

$$16\pi^2\,\mu\,\frac{dC_{12610}}{d\mu} = \beta^{(1)}_{C_{12610}}$$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1 y_2}}{MF}$
T2	$\frac{ye\overline{y_1 y_2 ye}}{MF}$
T3	$\frac{\sqrt{3}lambda12Inv2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{3g_2^2\overline{y_1 y_2}}{8MF}$	$-\frac{ye\overline{y_1 y_2 ye}}{8MF}$	$\frac{\sqrt{3}lambda12Inv2\overline{y_1 y_2}}{6MF}$

$\beta^{(1)}_{C_{12711}}$

Quantity: 1, 2, 7, 11

$$16\pi^2\,\mu\,\frac{dC_{12711}}{d\mu} = \beta^{(1)}_{C_{12711}}$$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1 y_2}}{MF}$
T2	$\frac{\sqrt{3}lambda12Inv2\overline{y_1 y_2}}{MF}$
T3	$\frac{ye\overline{y_1 y_2 ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_2^2\overline{y_1 y_2}}{8MF}$	$-\frac{\sqrt{3}lambda12Inv2\overline{y_1 y_2}}{6MF}$	$\frac{ye\overline{y_1 y_2 ye}}{8MF}$

$\beta^{(1)}_{C_{12712}}$

Quantity: 1, 2, 7, 12

$$16\pi^2\,\mu\,\frac{dC_{12712}}{d\mu} = \beta^{(1)}_{C_{12712}}$$

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1 y_2}}{MF}$
T2	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1 y_2}}{MF}$
T3	$\frac{iye\overline{y_1 y_2 ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3ig_2^2\overline{y_1 y_2}}{8MF}$	$-\frac{\sqrt{3}ilambda12Inv2\overline{y_1 y_2}}{6MF}$	$\frac{iye\overline{y_1 y_2 ye}}{8MF}$

$\beta^{(1)}_{C_{12811}}$

Quantity: 1, 2, 8, 11

$$16\pi^2\,\mu\,\frac{dC_{12811}}{d\mu} = \beta^{(1)}_{C_{12811}}$$

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1 y_2}}{MF}$
T2	$\frac{iye\overline{y_1 y_2 ye}}{MF}$
T3	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{3ig_2^2\overline{y_1 y_2}}{8MF}$	$-\frac{iye\overline{y_1 y_2 ye}}{8MF}$	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1 y_2}}{6MF}$

$\beta^{(1)}_{C_{12812}}$

Quantity: 1, 2, 8, 12

$$16\pi^2\,\mu\,\frac{dC_{12812}}{d\mu} = \beta^{(1)}_{C_{12812}}$$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1 y_2}}{MF}$
T2	$\frac{\sqrt{3}lambda12Inv2\overline{y_1 y_2}}{MF}$
T3	$\frac{ye\overline{y_1 y_2 ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_2^2\overline{y_1 y_2}}{8MF}$	$-\frac{\sqrt{3}lambda12Inv2\overline{y_1 y_2}}{6MF}$	$\frac{ye\overline{y_1 y_2 ye}}{8MF}$

$\beta^{(1)}_{C_{12911}}$

Quantity: 1, 2, 9, 11

$$16\pi^2\,\mu\,\frac{dC_{12911}}{d\mu} = \beta^{(1)}_{C_{12911}}$$

Column	Representative term structure
T1	$\frac{lambdaS2Inv1\overline{y_2}^2}{MF}$
T2	$\frac{g_2^2\overline{y_2}^2}{MF}$
T3	$\frac{ye\overline{y_2}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{lambdaS2Inv1\overline{y_2}^2}{4MF}$	$\frac{3g_2^2\overline{y_2}^2}{8MF}$	$-\frac{ye\overline{y_2}^2\overline{ye}}{8MF}$

$\beta^{(1)}_{C_{12912}}$

Quantity: 1, 2, 9, 12

$$16\pi^2\,\mu\,\frac{dC_{12912}}{d\mu} = \beta^{(1)}_{C_{12912}}$$

Column	Representative term structure
T1	$\frac{ilambdaS2Inv1\overline{y_2}^2}{MF}$
T2	$\frac{ig_2^2\overline{y_2}^2}{MF}$
T3	$\frac{iye\overline{y_2}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{ilambdaS2Inv1\overline{y_2}^2}{4MF}$	$\frac{3ig_2^2\overline{y_2}^2}{8MF}$	$-\frac{iye\overline{y_2}^2\overline{ye}}{8MF}$

$\beta^{(1)}_{C_{121011}}$

Quantity: 1, 2, 10, 11

$$16\pi^2\,\mu\frac{dC_{121011}}{d\mu} = \beta^{(1)}_{C_{121011}}$$

Column	Representative term structure
T1	$\frac{ilambdaS2Inv1\overline{y_2}^2}{MF}$
T2	$\frac{ig_2^2\overline{y_2}^2}{MF}$
T3	$\frac{iye\overline{y_2}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{ilambdaS2Inv1\overline{y_2}^2}{4MF}$	$\frac{3ig_2^2\overline{y_2}^2}{8MF}$	$-\frac{iye\overline{y_2}^2\overline{ye}}{8MF}$

$\beta^{(1)}_{C_{121012}}$

Quantity: 1, 2, 10, 12

$$16\pi^2\,\mu\frac{dC_{121012}}{d\mu} = \beta^{(1)}_{C_{121012}}$$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_2}^2}{MF}$
T2	$\frac{lambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{ye\overline{y_2}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_2^2\overline{y_2}^2}{8MF}$	$-\frac{lambdaS2Inv1\overline{y_2}^2}{4MF}$	$\frac{ye\overline{y_2}^2\overline{ye}}{8MF}$

$\beta^{(1)}_{C_{2211}}$

Quantity: 2, 2, 1, 1

$$16\pi^2\,\mu\frac{dC_{2211}}{d\mu} = \beta^{(1)}_{C_{2211}}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}lambdaT_3\overline{y_1}\,\overline{y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{2\sqrt{3}lambdaT_3\overline{y_1}\,\overline{y_2}}{3MF}$

$\beta^{(1)}_{C_{2212}}$

Quantity: 2, 2, 1, 2

$$16\pi^2\,\mu\frac{dC_{2212}}{d\mu} = \beta^{(1)}_{C_{2212}}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}ilambdaT_3\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{2\sqrt{3}ilambdaT_3\overline{y_1 y_2}}{3MF}$

$\beta^{(1)}_{C_{2222}}$

Quantity: 2, 2, 2, 2

$$16\pi^2\,\mu\frac{dC_{2222}}{d\mu} = \beta^{(1)}_{C_{2222}}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}lambdaT_3\overline{y_1}\,\overline{y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1}\,\overline{y_2}}{3MF}$

$\beta^{(1)}_{C_{2277}}$

Quantity: 2, 2, 7, 7

$$16\pi^2\,\mu\frac{dC_{2277}}{d\mu} = \beta^{(1)}_{C_{2277}}$$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1}^2}{MF}$
T2	$\frac{lambdaS1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{ye\overline{y_1}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_2^2\overline{y_1}^2}{4MF}$	$-\frac{lambdaS1Inv1\overline{y_1}^2}{2MF}$	$\frac{ye\overline{y_1}^2\overline{ye}}{4MF}$

$\beta^{(1)}_{C_{2278}}$

Quantity: 2, 2, 7, 8

$$16\pi^2\,\mu\frac{dC_{2278}}{d\mu} = \beta^{(1)}_{C_{2278}}$$

Column	Representative term structure
T1	$\frac{ilambdaS1Inv1\overline{y_1}^2}{MF}$
T2	$\frac{ig_2^2\overline{y_1}^2}{MF}$
T3	$\frac{iye\overline{y_1}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{ilambdaS1Inv1\overline{y_1}^2}{2MF}$	$\frac{3ig_2^2\overline{y_1}^2}{4MF}$	$-\frac{iye\overline{y_1}^2\overline{ye}}{4MF}$

$\beta^{(1)}_{C_{2279}}$

Quantity: 2, 2, 7, 9

$$16\pi^2\,\mu\frac{dC_{2279}}{d\mu} = \beta^{(1)}_{C_{2279}}$$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1 y_2}}{MF}$
T2	$\frac{ye\overline{y_1 y_2 ye}}{MF}$
T3	$\frac{\sqrt{3}lambda12Inv2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{3g_2^2\overline{y_1 y_2}}{4MF}$	$-\frac{ye\overline{y_1 y_2 ye}}{4MF}$	$\frac{\sqrt{3}lambda12Inv2\overline{y_1 y_2}}{3MF}$

$\beta^{(1)}_{C_{22710}}$

Quantity: 2, 2, 7, 10

$$16\pi^2\,\mu\,\frac{dC_{22710}}{d\mu} = \beta^{(1)}_{C_{22710}}$$

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1 y_2}}{MF}$
T2	$\frac{iye\overline{y_1 y_2 ye}}{MF}$
T3	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{3ig_2^2\overline{y_1 y_2}}{4MF}$	$-\frac{iye\overline{y_1 y_2 ye}}{4MF}$	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1 y_2}}{3MF}$

$\beta^{(1)}_{C_{2288}}$

Quantity: 2, 2, 8, 8

$$16\pi^2\,\mu\frac{dC_{2288}}{d\mu} = \beta^{(1)}_{C_{2288}}$$

Column	Representative term structure
T1	$\frac{lambdaS1Inv1\overline{y_1}^2}{MF}$
T2	$\frac{g_2^2\overline{y_1}^2}{MF}$
T3	$\frac{ye\overline{y_1}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{lambdaS1Inv1\overline{y_1}^2}{2MF}$	$\frac{3g_2^2\overline{y_1}^2}{4MF}$	$-\frac{ye\overline{y_1}^2\overline{ye}}{4MF}$

$\beta^{(1)}_{C_{2289}}$

Quantity: 2, 2, 8, 9

$$16\pi^2\,\mu\frac{dC_{2289}}{d\mu} = \beta^{(1)}_{C_{2289}}$$

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1 y_2}}{MF}$
T2	$\frac{\sqrt{3}ilambda12Inv2\overline{y_1 y_2}}{MF}$
T3	$\frac{iye\overline{y_1 y_2 ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3ig_2^2\overline{y_1 y_2}}{4MF}$	$-\frac{\sqrt{3}ilambda12Inv2\overline{y_1 y_2}}{3MF}$	$\frac{iye\overline{y_1 y_2 ye}}{4MF}$

$\beta^{(1)}_{C_{22810}}$

Quantity: 2, 2, 8, 10

$$16\pi^2\,\mu\,\frac{dC_{22810}}{d\mu} = \beta^{(1)}_{C_{22810}}$$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1 y_2}}{MF}$
T2	$\frac{ye\overline{y_1 y_2 ye}}{MF}$
T3	$\frac{\sqrt{3}lambda12Inv2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{3g_2^2\overline{y_1 y_2}}{4MF}$	$-\frac{ye\overline{y_1 y_2 ye}}{4MF}$	$\frac{\sqrt{3}lambda12Inv2\overline{y_1 y_2}}{3MF}$

$\beta^{(1)}_{C_{2299}}$

Quantity: 2, 2, 9, 9

$$16\pi^2\,\mu\frac{dC_{2299}}{d\mu} = \beta^{(1)}_{C_{2299}}$$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_2}^2}{MF}$
T2	$\frac{lambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{ye\overline{y_2}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_2^2\overline{y_2}^2}{4MF}$	$-\frac{lambdaS2Inv1\overline{y_2}^2}{2MF}$	$\frac{ye\overline{y_2}^2\overline{ye}}{4MF}$

$\beta^{(1)}_{C_{22910}}$

Quantity: 2, 2, 9, 10

$$16\pi^2\,\mu\,\frac{dC_{22910}}{d\mu} = \beta^{(1)}_{C_{22910}}$$

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_2}^2}{MF}$
T2	$\frac{ilambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{iye\overline{y_2}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3ig_2^2\overline{y_2}^2}{4MF}$	$-\frac{ilambdaS2Inv1\overline{y_2}^2}{2MF}$	$\frac{iye\overline{y_2}^2\overline{ye}}{4MF}$

$\beta^{(1)}_{C_{221010}}$

Quantity: 2, 2, 10, 10

$$16\pi^2\,\mu\frac{dC_{221010}}{d\mu} = \beta^{(1)}_{C_{221010}}$$

Column	Representative term structure
T1	$\frac{lambdaS2Inv1\overline{y_2}^2}{MF}$
T2	$\frac{g_2^2\overline{y_2}^2}{MF}$
T3	$\frac{ye\overline{y_2}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{lambdaS2Inv1\overline{y_2}^2}{2MF}$	$\frac{3g_2^2\overline{y_2}^2}{4MF}$	$-\frac{ye\overline{y_2}^2\overline{ye}}{4MF}$